

Comments to FDA on Guidelines for HCT/P

I oppose the finalization of the current draft guidelines for HCT/P.

I have been in private surgical practice as a solo practitioner since 1977. I work in a AAAHC certified facility and track our cases in accordance with AAAHC requirements. I became aware of the utility of Platelet rich plasma and supravascular fraction cells extracted from patient's fat in 2013 as a result of the continuing education provided by the American Society of Cosmetic Surgery.

I attended a training course provided by Bioheart Inc (US Stem Cell Inc.) in 2013. That technology was incorporated into my cosmetic surgery practice (approximately 175 surgical cases per year) in October 2013 and has resulted in positive improvement in outcomes in all cases. There have been 544 uses of PRP (Platelet Rich Plasma) in surgical and other cases in my practice since October 2013. There have been no adverse outcomes attributed to the use of PRP. We have used ADSVF (Adipose derived supravascular fraction) material in 25 cases to enhance fat grafting. Results have been positive with no adverse outcomes. All materials are prepared by our licensed trained personnel in a sterile environment using CGMP (Current good manufacturing practice) technique using protocols provided by US Stem Cell, Inc. We obtain separate informed consent for the use of these materials from each patient and provide written information for them prior to the surgical case. A checklist CGMP document is prepared for every case.

Our surgical adverse outcome rate has dropped by 50% or more in every class of procedure we perform. This includes major breast surgery, abdominal contouring surgery, hernia repair, liposuction, nasal reconstructive surgery, fat grafting to the body and face and facial rejuvenation procedures. We have begun using the PRP applied topically following Chemical peels and laser treatment. I have been able to salvage two exposed breast implants using this material. Post operative scar treatment with PRP is dramatically more successful than with standard techniques. In every instance, the results are beyond our expectations and adverse outcomes are reduced.

I was able to integrate this technology into my practice because of our ability to create the PRP and ADSVF onsite from the patient at the same surgical procedure. The increase in cost to the patients has been minimal and reflects the cost of supplies for the case.

I became aware of the FDA hearing on HCT/P (Human Cells, Tissues, and Cellular and Tissue-Based Products) late in 2015 after notification from our support company, US Stem Cell. I attended the FDA Hearing on Sept 12, 2016 for the entire day and was able to view the September 13, 2016 testimony online.

In review of the comments from prior hearings, the FDA uses adjectives such as "Some, Many and a few". I find it curious that specific numbers and their relationship to the total number of comments were not used. This informal reporting methodology is contrary to the specific, statistical data required by the FDA for evaluation of Drugs and medical equipment. I urge your staff to use specific numbers of comments and their relative percentage rates by topic when reporting the commentary for this hearing.

The 4 draft guidelines were created a number of years ago. There may have been some scientific or other logical basis for some of the guidelines when they were created. However, the advancement of knowledge since the initial publication of the guidelines has changed dramatically. I urge the FDA to avoid the use and enforcement of guideline recommendations that today, are arbitrary and capricious and not based on science.

Examples of these arbitrary and capricious guidelines are contained in the homologous use guideline. One of the examples used as non structural use of cellular material is the injection of fat into the breast. The example used is that the breast is primarily a lactating organ and that fat does not satisfy this organ function. It should be noted that the breast is made up of both fat and breast tissue and that injection of fat into the breast is absolutely within the appropriate uses of that material. I routinely use PRP to enhance fat grafting for breast reconstructive procedures following mastectomy and have found it to be safe and effective.

Other examples of nonstructural use of tissues not mentioned in the guidelines includes red blood cells used as a blood patch for leaking spinal fluid, the use of red blood cells and red blood cell clot in dental extractions to help fill a defect in the bone after tooth extraction, the use of platelet rich plasma and fibrin injected into bone grafts to assist in healing, the injection of fat cells beneath wrinkles to smooth the wrinkle. (This involves placing fat into dermal tissue, a nonstructural use according to the definition offered by the FDA)

The draft guidelines are exceptionally intrusive into my practice of medicine. The FDA has excluded PRP from guidance. This is curiously inconsistent since Platelet rich

plasma extraction involves the use of a chemical (sodium citrate) to significantly alter the character of blood so that centrifuge separation can occur. In addition, scientific literature acknowledgement of the effectiveness of PRP is still pending. The FDA draft guidelines, if finalized, will definitely interfere with my ability to perform fat grafting and other procedures.

Medical practitioners are in the unique position to understand the specific needs of their patients and have historically utilized medications, tissue and other substances in an off label basis when appropriate. By interfering with this ability, the FDA is directly intruding into the clinical practice of medicine of the individual practitioner and interfering in the relationship between the clinician and the patient. It seems inappropriate that the FDA should intrude on this long recognized practice. The practitioner and the patient should be able to make informed decisions about care without unnecessary intervention from your federal agency.

I understand the concern about fraudulent claims and patient injury. There were several patients who testified at your recent hearing in support of stem cell treatment. I find it interesting that there were no questions from the panel for any of these participants. In one case, the patient had stem cells created in the US but had to travel to Mexico to have them implanted. I would have thought your panel members would have been very interested in the circumstances of that treatment.

Throughout history, some in the organized medical community resist change that seems contrary to conventional wisdom. I recall when I opened my medical practice in 1977; a local businessman tried to donate a hyperbaric chamber to the local hospital but was denied because the medical staff felt it was quackery. In another case, the medical staff did not want to restrict patient smoking in the hospital because there was not real proof that smoking was harmful to health and they were afraid we would lose patients to other areas.

I suspect we are revisiting a “Simmelweis effect” with regard to the use of PRP and Stem cells. I urge the FDA to exercise caution in finalizing any guidelines that do not have scientific basis and which are intrusive into the physician patient relationship.

There are numerous ways to protect the public from fraudulent practitioners and improve our knowledge in this field.

- While there are some organizations that are collecting data for their members, an FDA Registry could be formed to collect large amounts of real time data for the benefit of all medical practitioners and the public. This is currently being

done with opioid use, CME requirements, databank reporting, etc. This public information could include the names of cooperating facilities and physicians. Annual (or more frequent) review and updates of treatment outcomes that are available in a national registry could be very valuable to legitimate practitioners.

- Content Analysis of these new biologic products is expensive and not widely available. The FDA's resources could assist practitioners to determine the content of these products in a cost effective, consistent manner. AS an example, I can easily order a CBC and chemistry profile, but have difficulty getting a platelet count on PRP or a stem cell count and analysis on ADSVF. Widely available, cost effective content analysis of biologic materials would go a long way to assist those of us in private practice to utilize these treatments effectively.
- For same day surgical procedures, the FDA could regulate the preparation protocols of these cellular materials (including guidelines for content analysis) not the materials themselves. Guidelines for the safe and sterile preparation of biologic material in a same surgical procedure could be made public. With knowledge of such guidelines, facilities could meet known recommendations just as they currently meet other safety guidelines (ie. Surgical site identification, surgical check list, time out in OR, Surgical Safety, experimental use, etc.). Introduction of safety guidelines has been most successful when they are part of the facility standard. There are several facility accreditation organizations, (AAAHC, JCAHO, AAAAHC) that could make reasonable sterility and CGMP guidelines a standard for the majority of health care facilities within a short period of time.

Biologic medical and surgical practice has progressed in our generation because of innovative uses of cellular materials by individual practitioners. I am seriously concerned that the FDA's proposed arbitrary, capricious and non scientific approach to regulation of cellular therapy will dramatically inhibit the utilization of these materials, perhaps in ways that today we can't even imagine. Certainly, the inability to obtain these treatments in our country will continue to drive patients outside the USA.

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